Shared sources | Warm-up and checkpoint

Topics 7-8: mechanisms checkpoint

Name ____________________________ Date _______________

All numbers are constructed classroom records. Warm-up letters M/N are separate from checkpoint letters A/B and U/V.

Warm-up only: M has 60 beats/min and 50 mL/beat; N has 80 beats/min and 50 mL/beat. W1 calculate N. W2 test “all hormones affect every cell equally”. W3 classify a pulmonary artery using direction.

Checkpoint pump	Heart rate (beats/min)	Stroke volume (mL/beat)
A	75	80
B	100	50

Pump output = heart rate × stroke volume. 1,000 mL = 1 L. The table does not measure health or fitness.

Exchange case	Relative area	Relative concentration difference	Relative thickness
U	4	3	1
V	4	3	2

Idealised relative index = area × concentration difference ÷ thickness, with the same omitted constant. These are model inputs, not measured human gas-exchange data.

Q1 context: glucose absorption has raised blood glucose. Q2 starts with oxygenated blood leaving the lungs. Q4 concerns quiet inhalation and oxygen exchange into arriving blood.

Supported checkpoint | 20 core marks

Same concepts and marks. Optional word bank: pancreas; liver; insulin; glycogen; atrium; ventricle; aorta. Record support used.

Q1 [4 marks]. Blood glucose has risen after glucose absorption. Name the gland and hormone involved, describe one target-tissue response, and state its effect on blood glucose. Use four short points.

Q2 [4 marks]. Complete: lungs → [chamber] → [chamber] → [large artery] → body. Then define an artery by direction.

Q3 [4 marks]. Use record A. Show the pump-output calculation in mL/min, convert to L/min, and explain why heart rate alone cannot always compare output.

Q4 [4 marks]. During inhalation, state the volume change and pressure change. Explain the direction of oxygen diffusion at an alveolus and why a thin barrier helps. Link each feature to its effect.

Q5 [4 marks]. Compare U and V: identify the changed factor, calculate V's index, compare V's index with U's, and state one limit of this model.

Standard checkpoint | 20 core marks

Complete Q1-Q5 using the shared sources. Show working and units.

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Q3 [4 marks]. Use record A. Show the pump-output calculation in mL/min, convert to L/min, and explain why heart rate alone cannot always compare output.

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Q5 [4 marks]. Compare U and V: identify the changed factor, calculate V's index, compare V's index with U's, and state one limit of this model.

Challenge checkpoint | Same 20 core marks

Complete the same core first. The final audit is optional and unscored.

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Q4 [4 marks]. During inhalation, state the volume change and pressure change. Explain the direction of oxygen diffusion at an alveolus and why a thin barrier helps.

Q5 [4 marks]. Compare U and V: identify the changed factor, calculate V's index, compare V's index with U's, and state one limit of this model.

Optional audit: explain what additional evidence you would need before using pump records to judge whether oxygen supply meets a tissue's demand. Use spare paper or an oral response; no extra marks.

Feedback and next step | Keep the first attempt

Complete after the independent checkpoint. This is a selected-topic check, not a GCSE grade.

Record: route used; calculator; read-aloud; any content prompts or additional support.

Item	Focus	Mark /4	One next step
Q1	Insulin response		
Q2	Route and vessel definition		
Q3	Pump calculation and units		
Q4	Ventilation and exchange		
Q5	Model comparison and limit		

Total /20: ____ . L1. Preserve your first answer; write one correction and explain the scientific reason.

E1. One idea I can explain now. E2. One precise practice for my next attempt.